

Diptesh Das

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About Me

Computer Science undergraduate specializing in LLM systems, autonomous agents, multimodal AI, computer vision, and geospatial machine learning. Experienced in building production-oriented AI applications, reliable backend services, and end-to-end ML pipelines from research to deployment.

Education

Vellore Institute of Technology (VIT), Vellore 2024–Present
B.Tech in Computer Science and Engineering; CGPA: 9.18

Indian Institute of Technology (IIT) Madras 2024–Present
B.S. in Data Science and Applications; CGPA: 8.4

Experience

Founding AI Engineer, nRouter (California, USA) 2026–Present

- Building secure enterprise AI infrastructure for multi-model routing, governance, budget enforcement and unified access to leading LLM providers.
- Developing evaluation harnesses and agent integrations into open-source systems like Hermes and Openclaw to compare model latency, cost, reliability, fallback behaviour and output quality.

Research Intern, Samsung PRISM R&D Program 2026–Present

- Developing autonomous browser agents that plan and execute multi-step tasks across dynamic websites with minimal human intervention.
- Building DOM-aware retrieval and structured site representations using browser automation, vector search, graph databases and ontologies.
- Designing adaptive retry, memory and interaction pipelines to improve agent robustness across changing page structures and AMP websites.

Research Intern, IIT Kharagpur 2025

- Engineered a spatio-temporal ML pipeline over **traffic-accident records** using Google Earth Engine and satellite-derived features.
- Performed geospatial feature engineering, temporal analysis, hotspot identification and predictive accident-risk modelling.

AI Engineer, Valsco Technology (Jurident AI) 2026

- Engineered a legal AI assistant for a platform serving **4k+ users**, using Vectorless RAG, query rewriting and LLM orchestration.
- Improved retrieval through phrase matching, identifier boosting, hybrid search, conversational memory and citation-aware legal responses.
- Built production FastAPI services for legal research, structured drafting, document analysis and export-ready Word/PDF outputs.

AI Researcher, Autonomous Vehicle Research Centre, VIT Jan–Jul 2026

- Developed swarm-intelligence algorithms for Crazyflie drones, improving multi-agent coordination efficiency by **25%**.
- Contributed to computer-vision pipelines for road and building extraction from aerial imagery with **98% detection accuracy**.
- Worked on EVGRID AI - autonomous navigation, path planning, perception and coordination workflows for electric vehicles.

Projects

Turbulence – LLM Reliability Evaluation Platform / *FastAPI, SSE, LLM APIs* [GitHub](#)

- Built an AI reliability platform that stress-tests LLM applications under provider outages, latency spikes, malformed responses and routing failures.
- Developed reproducible scenarios to compare fallback behaviour, response quality, latency and resilience across model providers.
- Implemented an SSE-based monitoring cockpit and validated the evaluation workflow through **15 passing test fixtures**.

MindCanvas – AI Visual Journaling Platform / *FastAPI, React, GenAI* [GitHub](#)

- Developed a multimodal GenAI platform that converts short journal entries into personalized comic-style visual narratives.
- Combined emotion classification, LLM-based story generation, structured prompting, and image synthesis in an end-to-end pipeline..

XyraChain – Explainable AI Diagnostics Platform / *Deep Learning, Grad-CAM, FastAPI* [GitHub](#)

- Built an explainable AI system for chest X-ray classification using deep-learning models and Grad-CAM attention visualizations.
- Developed FastAPI inference services for image processing, prediction generation and structured diagnostic reporting.

Spatio-Temporal Accident Risk Prediction / *ML, GEE, Geospatial Analytics* [IEEE Xplore](#)

- Developed a geospatial ML framework over **official accident records**, combining traffic, temporal, environmental and satellite-derived features.
- Used Google Earth Engine for feature extraction, hotspot analysis and spatio-temporal accident-risk forecasting.
- Achieved an R^2 of approximately **0.88**, resulting in a peer-reviewed IEEE InGARSS publication.

Leadership & Positions

Team Lead, Team Prometheus (RoboSoccer): Led multi-agent robotics, path-planning integration, and autonomous coordination.

Senior Core, CSI-VIT: Led AI initiatives, technical workshops, peer mentoring, and developer-community activities.

Technical Skills

Programming: Python, SQL, Java, C++, TypeScript, JavaScript, Solidity

AI/ML: Machine Learning, Deep Learning, LLMs, RAG, NLP, Computer Vision, Multi-Agent Systems, Swarm Intelligence, Geospatial ML

Frameworks/Tools: PyTorch, TensorFlow, Keras, FastAPI, OpenCV, Google Earth Engine, CrewAI, Power BI, Databricks

Databases/Infrastructure: PostgreSQL, SQLite FTS5, Docker, Git, Linux, REST APIs, Google Colab

Certifications

Microsoft Power BI: Data Analytics & ETL, 2024 | **IBM:** Web Development, 2025